SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

ESS BMS Controller V3 (ESS-BMS-V3) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: 1,000 mm in front for maintenance; ambient controlled by HVAC between 15 and 30 °C
- Operating temperature: -10 to +50 °C
- Degree of protection: IP20
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: see dimension drawing
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Check battery rack insulation resistance
2. Verify CAN communication between PCS and BMS
3. Select grid code profile (parameter G-01)
4. Perform pre-charge test
5. Run charge/discharge test at 20% power

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See
DC insulation fault	Moisture in battery rack DC cabling	TG-ESS-BMS-V3 §3.1
BMS communication loss on PCS	Firmware compatibility: BMS 3.4 heartbeat vs PCS < 2.62	TG-ESS-BMS-V3 §3.2
Intermittent CAN errors	CAN termination missing	TG-ESS-BMS-V3 §3.3
Cell over-temperature	Container HVAC failure	TG-ESS-BMS-V3 §3.5

On-site commissioning support can be requested through the Global CS Center. For energy storage systems the request is handled by the technical support team.